

Table 4: Comparison of Powerful Open-source and Closed-source LLMs

Model	Pre-release Accuracy	First Post-release Accuracy	Decline	Percentage Decline
<u>Baichuan-13B-Chat</u>	0.52	0.23	0.29	55.77%
GPT-3.5-turbo-230613	0.49	0.33	0.16	32.65%
GPT-4-230613	0.60	0.41	0.19	31.67%
Llama-2-13B	0.52	0.45	0.07	12.50%
Llama-2-7B	0.27	0.27	0.01	1.85%
LLaMA2-7B-Chat	0.45	0.38	0.07	15.56%
Baichuan2-13B-Base	0.49	0.38	0.11	22.45%
Baichuan2-13B-Chat	0.50	0.38	0.12	24.00%
Baichuan2-7B-Base	0.42	0.28	0.14	33.33%
Baichuan2-7B-Chat	0.37	0.39	-0.02	-5.41%
Colossal-LLaMA-2-7B-Base	0.42	0.40	0.02	4.76%
Mistral-7B-v0.1	0.44	0.38	0.06	13.64%
Phi-1.5	0.41	0.38	0.03	7.32%
Qwen-14B-Chat	0.39	0.24	0.15	37.18%
Zephyr-7B-beta	0.40	0.32	0.08	20.00%
Gemini	0.43	0.29	0.14	33.72%
Yi-6B	0.29	0.22	0.07	25.86%
Qwen-1.8B	0.35	0.29	0.05	15.71%
GPT-4-231106	0.66	0.42	0.24	36.36%
Phi-2	0.31	0.23	0.08	27.42%
Command R+	0.54	0.40	0.14	25.93%
Claude-3-opus-20240229	0.53	0.45	0.09	16.04%
Claude-3-sonnet-20240229	0.23	0.18	0.05	19.57%
DeepSeek-V2-Chat	0.58	0.47	0.10	18.10%
Mixtral-8x22B-Instruct-v0.1	0.56	0.41	0.15	26.79%
Phi-3-mini-4k-Instruct	0.38	0.25	0.13	34.21%
Qwen1.5-110B-Chat	0.56	0.43	0.14	24.11%
<u>Claude-3.5-Sonnet-20240620</u>	0.63	0.35	0.28	43.65%
Gemini-1.5-Pro	0.50	0.41	0.09	18.00%
<u>Gemini-1.5-flash</u>	0.39	0.23	0.17	42.31%
Qwen2-72B-Instruct	0.60	0.43	0.17	29.17%
LLaMA-3.1-405B-Instruct	0.50	0.41	0.09	18.00%
GPT-4o	0.62	0.39	0.23	37.10%
<u>GPT-4o-mini-2024-07-18</u>	0.48	0.28	0.20	41.67%

- **Question and Answer Forums:** Quora, focusing on popular topics such as Technology, Mathematics, Health, and Movies.

To facilitate efficient and effective data collection, our system’s environment is configured with a specific Python package index URL and utilizes Playwright for web navigation and content extraction. This setup addresses potential challenges, such as page refresh requirements on Yahoo and hostname verification issues encountered with cer-

tain websites.

Our system employs a combination of tools and libraries, including Requests, Playwright, and various PDF processing utilities, to collect the latest textual data from the internet. This approach enables us to dynamically update our benchmark datasets with fresh information, ensuring that our evaluation reflects current language use and information trends.

Data Pre-processing For some raw content

clean the text by removing these tags to ensure that only the content text is analyzed.

Filtering out short texts: Extremely short texts may not provide enough contextual information for analysis or model training. Setting a minimum text length threshold of 100 chars to filter out such instances.

External Dataset For the analysis of text likelihood and bias, we employed not only our own collection of historical arXiv data but also sampled entries from two additional datasets. We incorporated data from the BBC News and Wikitext datasets, which are publicly available through the Hugging Face datasets repository. Specifically, the datasets used were *bbc_news_alltime* and *wikitext_alltime*, accessible via the following URLs:

- BBC News: https://huggingface.co/datasets/RealTimeData/bbc_news_alltime
- Wikitext: https://huggingface.co/datasets/RealTimeData/wikitext_alltime

Sampling from these datasets was carried out every three months, with each instance involving the collection of 50 entries per dataset. This structured sampling approach was designed to ensure a consistent and representative analysis of content over time, enabling an effective assessment of textual bias and likelihood trends.

G Data Overview of One Crawl

Tab. 9 presented below offers a comprehensive overview of the diverse sources and categories of texts that were analyzed in one data crawl. Classified into distinct groups such as Academic STEM, Academic Non-STEM, Internet QA, and Internet articles, the table delineates the number of texts extracted from each source along with their average length. This includes detailed counts from different academic archives like arXiv, various categories from Internet question and answer platforms like Reddit and Quora, and articles from well-known online platforms including Wiki and BBC.

H Detailed results of BPC generalization on Wiki and BBC dataset

See Table 6.

Table 5: Most common tags of the Gjo questions

Topic	Count
Business	648
Non-US Politics	453
Security and Conflict	446
Society	425
US Politics	363
Technology	359
Finance	299
Elections and Referenda	288
Economic Indicators	273
Foreign Policy	271
US Policy	247
Health	245
Leader Entry/Exit	219
Economic Policy	160
Sports	148
Entertainment	148
Environment	80
Open	16

I TBI of Models across the arXiv, BBC, and Wiki datasets

See Table 8.

J Gjo data samples and most common tags of the Gjo questions

Gjo data samples see Table 5 and Gjo questions see Table 10.

K Analysis of Language Likelihood Values and Benchmark Correlation Across Text and Model Groups

From Fig. 12 we can conclude that the scale of models being observed does not significantly alter outcomes, suggesting that performance correlations across benchmarks are generally consistent regardless of model size.

L Correlation with Existing Benchmarks

The Selected Existing Benchmarks We explored the correlation between BPC and existing benchmarks, particularly focusing on how specific content types, such as GSM, influence model performance. The selected benchmarks encompass common sense reasoning, depth and breadth of knowledge across diverse subjects, truthfulness, natural language understanding, and mathematical

Table 6: Percentage Change in BPC Following Release, measured by BBC news data from 2020 to 2024.3. Dash ("-") symbols in the table indicate missing data points due to the later release dates of some models, which means there is no available data for those specific time intervals.

Model	Base BPC	BPC Change at 3 months (%)	BPC Change at 6 months (%)	BPC Change at 9 months (%)	BPC Change at 12 months (%)	mean
OPT-13B	0.450	2.273	-2.087	1.265	0.839	0.572
OPT-2.7B	0.475	1.970	-2.228	1.192	1.143	0.519
LLaMA-7B	0.450	1.270	0.177	-7.776	3.266	-0.766
Pythia-12B	0.480	-0.233	-3.888	-2.612	-1.012	-1.936
Falcon-rw-1B	0.510	0.459	-7.814	1.201	-1.474	-1.907
Baichuan-13B-Chat	0.467	-3.645	-1.078	0.413	-	-1.436
LLaMA-2-13B	0.416	-6.928	3.502	1.683	-	-0.581
LLaMA-2-7B	0.434	-7.314	3.092	1.279	-	-0.981
LLaMA-2-7B-Chat	0.505	-8.308	2.695	0.664	-	-1.650
Zhongjing-Base	0.465	-8.488	2.662	-1.290	-	-2.372
InternLM-Chat-7B	0.515	-2.476	-1.344	0.236	-	-1.195
Baichuan2-7B-Base	0.451	-3.059	0.595	-4.649	-	-2.371
Baichuan2-7B-Chat	0.517	-3.571	-0.038	-5.315	-	-2.975
Mistral-7B-v0.1	0.398	1.110	5.347	-0.661	-	1.932
Phi-1.5	0.602	-1.772	1.131	-1.837	-	-0.826
Baichuan2-13B-Base	0.439	-2.943	0.882	-4.390	-	-2.150
Baichuan2-13B-Chat	0.495	-3.541	-0.995	0.232	-	-1.435
Colossal-LLaMA-2-7B-Base	0.618	2.825	3.715	-	-	3.270
Qwen-14B-Chat	0.445	2.682	4.847	-	-	3.764
Qwen-7B	0.441	1.892	3.689	-	-	2.791
Qwen-7B-Chat	0.466	1.809	3.606	-	-	2.708
Skywork-13B-Base	0.429	7.163	3.261	-	-	5.212
ChatGLM3-6B	0.832	5.567	1.684	-	-	3.626
Zephyr-7B-beta	0.435	7.340	3.928	-	-	5.634
Yi-6B	0.427	6.478	3.144	-	-	4.811
Yi-6B-Chat	0.446	3.030	-2.798	-	-	0.116
Qwen-1.8B	0.520	1.899	-2.082	-	-	-0.091
Qwen-1.8B-Chat	0.599	1.635	-2.033	-	-	-0.199
RWKV-v5-Eagle-7B	0.440	4.956	2.278	-	-	3.617
TinyLLaMA-1.1B-Chat-v0.6	0.513	2.539	-3.678	-	-	-0.569
Phi-2	0.487	1.882	-	-	-	1.882

reasoning, ensuring a well-rounded assessment of the model’s performance.

Base vs. Chat Models Tab. 12 shows the comparative analysis of base and tuned models. Our findings suggest that: 1) Base models exhibit a stronger correlation with established benchmarks, implying that these models, in their fundamental form, possess a robust capacity for modeling language. This observation advocates for conducting initial evaluations using base models to capture their intrinsic language modeling capabilities. 2) The exploration of tuned models reveals potential deeper patterns in model tuning, suggesting that specific tuning strategies may unlock further enhancements in model performance across different tasks and domains.

Correlation alone Text Length Tab. 13 is about a long text understanding benchmark Longbench.

For chat models in this task, the correlation be-

tween BPC and scores is logical—higher probabilities of generating the text correlate with higher scores, and this correlation strengthens with longer text lengths. This suggests a consistent alignment between the model’s language likelihood in long text and its scoring, reinforcing the chat model’s robustness in handling extensive text inputs.

However, an inverse trend is observed with base models, where a familiar text paradoxically results in lower scores on related comprehension tasks. This indicates that fine-tuning may alter the base models’ understanding of the texts, perhaps shifting their processing in ways that do not favor traditional comprehension metrics.

This indicates scenarios where a model scores high on language likelihood due to its text-generation skills but may not perform equally well on tasks that require deep semantic understanding or critical thinking. Thus, relying solely on language likelihood as an indicator of overall per-